Sulabham · APPSC AEE 2026 — Module 3 Test: Surveying

Paper 3 Civil · original practice questions in the pattern of APPSC AEE Paper III Civil, 22 Aug 2023 · take at the end of Week 7

Time: 50 minutes (exam pace: 1 minute per question) **Marks:** +1 correct, −⅓ wrong, 0 unattempted
Calculator: not allowed
Write your answers on a separate sheet. Mark a question only if you can rule out at least two options.
Online version with timer: sulabham.in/AEE_Module3_Test.html

Questions

A. Fundamentals, Scales and Errors

1. Assertion (A): In plane surveying the curvature of the earth is neglected.
Reason (R): For small areas, the difference between the length of an arc on the earth's surface and its chord is negligible.
(a) Both A and R are true and R is the correct explanation of A
(b) Both A and R are true, but R is not the correct explanation of A
(c) A is true, but R is false
(d) Both A and R are false

2. A plan drawn to a scale of 1 cm = 20 m has shrunk so that a line originally 10 cm long now measures 9.6 cm. An area measured on the shrunk plan is 92.16 cm². The true area on the ground is:
(a) 3.69 hectares (b) 3.84 hectares (c) 4.00 hectares (d) 4.34 hectares

3. Match List-I (type of survey) with List-II (its concern):

List-I	List-II
A. Cadastral survey	1. Rough preliminary inspection of the area
B. Hydrographic survey	2. Boundaries of properties
C. Geodetic survey	3. Curvature of the earth is taken into account
D. Reconnaissance survey	4. Water bodies and the bed below them

(a) A-2, B-4, C-3, D-1 (b) A-4, B-2, C-3, D-1 (c) A-2, B-3, C-4, D-1 (d) A-1, B-4, C-3, D-2

4. On a map of scale 1 : 25,000, a reservoir measures 16 cm². Its area on the ground is:
(a) 0.4 km² (b) 4 km² (c) 10 km² (d) 1 km²

B. Chain and Tape

5. A field was measured with a 30 m chain and its area was computed as 10,000 m². The chain was later found to be 15 cm too short. The true area of the field is:
(a) 10,100.25 m² (b) 9,950.00 m² (c) 9,900.25 m² (d) 10,050.00 m²

6. A 30 m steel tape weighing 12 N in all is supported only at its ends under a pull of 100 N. The sag correction is:
(a) 0.0018 m (b) 0.018 m (c) 0.036 m (d) 0.180 m

7. A 30 m steel tape standardised at 20 °C was used at 40 °C to measure a line, which was recorded as 600 m. If $\alpha = 12 \times 10^{-6}$ per °C, the true length of the line is:
(a) 599.856 m (b) 600.072 m (c) 600.144 m (d) 600.288 m

8. Study the statement and the conclusions and select the most appropriate option.
Statement: A line was measured with a 20 m chain which was later found to be 5 cm too long.
Conclusion 1: The measured length of the line is less than its true length.
Conclusion 2: The correction to be applied to the measured length is negative.
(a) Only conclusion 1 is correct
(b) Only conclusion 2 is correct
(c) Both conclusions 1 and 2 are correct
(d) Neither conclusion 1 nor 2 is correct

9. S1: Normal tension is the pull at which the correction for pull and the correction for sag neutralise each other.
S2: An invar tape is preferred for base-line measurement because of its high coefficient of thermal expansion.
(a) Both S1 and S2 are true (b) S1 is true and S2 is false (c) S1 is false and S2 is true (d) Both S1 and S2 are false

10. Which of the following is an obstacle to chaining but not to ranging?

(a) A hill between the two stations (b) A pond on the chain line (c) A building on the chain line (d) A dense forest on the chain line

C. Compass Surveying

11. The whole circle bearings of lines AB and BC are 75° and 160°. The interior included angle ABC (less than 180°) is:

(a) 85° (b) 235° (c) 265° (d) 95°

12. In a compass traverse: line PQ has FB 130°30′ and BB 310°30′; line QR has FB 200°15′ and BB 21°45′; the FB of line RS observed at R is 285°00′. The corrected fore bearing of RS is:

(a) 286°30′ (b) 285°00′ (c) 284°15′ (d) 283°30′

13. The true bearing of a line is 210° and the magnetic declination at the place is 4° W. The magnetic bearing of the line is:

(a) 214° (b) 206° (c) 210° (d) 218°

14. Match List-I with List-II:

List-I	List-II
A. Isogonic line	1. Zero dip
B. Agonic line	2. Equal declination
C. Isoclinic line	3. Equal dip
D. Aclinic line	4. Zero declination

(a) A-3, B-1, C-2, D-4 (b) A-2, B-1, C-3, D-4 (c) A-2, B-4, C-3, D-1 (d) A-4, B-2, C-1, D-3

15. Assertion (A): In a prismatic compass, sighting the object and reading the bearing are done together from one position.

Reason (R): The graduated ring of a prismatic compass is fixed to the box and turns with the line of sight.

(a) Both A and R are true and R is the correct explanation of A (b) Both A and R are true, but R is not the correct explanation of A

(c) A is true, but R is false (d) Both A and R are false

D. Theodolite, Traversing and Coordinates

16. Station A has coordinates N 200 m, E 100 m. Line AB is 100 m long with a bearing of S 36°52′ E (cos 36°52′ = 0.8, sin 36°52′ = 0.6). The coordinates of B are:

(a) N 280 m, E 160 m (b) N 120 m, E 160 m (c) N 140 m, E 180 m (d) N 120 m, E 40 m

17. A closed traverse has a perimeter of 800 m and the total error in latitude is +0.40 m. By Bowditch's rule, the correction to the latitude of a side 200 m long is:

(a) +0.10 m (b) −0.05 m (c) −0.40 m (d) −0.10 m

18. The coordinates of A are N 500 m, E 300 m and those of B are N 400 m, E 400 m. The whole circle bearing of the line AB is:

(a) 45° (b) 315° (c) 135° (d) 225°

19. Consider the following statements:

I. In a closed traverse the algebraic sum of the deflection angles is 360°.

II. The sum of the exterior angles of a closed traverse of n sides is (2n + 4) × 90°.

III. In an open traverse the closing error is found from the sums of latitudes and departures.

IV. The transit rule is used when the angular measurements are more precise than the linear measurements.

Which of the above statements are correct?

(a) I and II only (b) I, II and IV only (c) II, III and IV only (d) I, II, III and IV

20. Turning the telescope of a theodolite about its horizontal axis through 180° in the vertical plane is called:

(a) swinging the telescope (b) transiting the telescope (c) centring (d) changing face

21. Study the statement and the conclusions and select the most appropriate option.
Statement: A horizontal angle is measured with a transit theodolite on both faces, reading both verniers.
Conclusion 1: The error due to eccentricity of the verniers is eliminated by reading both verniers.
Conclusion 2: The error due to the vertical axis not being truly vertical is eliminated by taking both-face observations.
(a) Only conclusion 1 is correct (b) Only conclusion 2 is correct
(c) Both conclusions 1 and 2 are correct (d) Neither conclusion 1 nor 2 is correct

E. Levelling

22. From one set-up of a level the staff readings were 1.250 m on a bench mark of RL 50.000 m, then 1.875 m, and finally 0.940 m on point C. The RL of C is:
(a) 50.310 m (b) 49.690 m (c) 50.935 m (d) 51.250 m

23. The back sight on a bench mark of RL 100.000 m is 1.500 m. The staff is then held inverted against the underside of a beam and reads 2.500 m. The RL of the underside of the beam is:
(a) 99.000 m (b) 101.500 m (c) 103.000 m (d) 104.000 m

24. In a levelling operation $\Sigma BS = 6.520$ m and $\Sigma FS = 4.870$ m. If the RL of the first point is 150.000 m, the RL of the last point is:
(a) 148.350 m (b) 151.560 m (c) 161.390 m (d) 151.650 m

25. A lighthouse lamp is 64 m above sea level. An observer's eye on a ship is 9 m above sea level. Using $d = 3.855\sqrt{h}$ (d in km, h in m), the greatest distance at which the lamp is just visible is about:
(a) 42.4 km (b) 32.9 km (c) 30.8 km (d) 11.6 km

26. A staff is held 100 m from a level. When the bubble is moved through 5 divisions, the staff reading changes by 0.050 m. The angular value of one division of the bubble tube is about (1 radian = 206,265″):
(a) 10.3″ (b) 20.6″ (c) 41.2″ (d) 103″

27. Match List-I (type of levelling) with List-II (description):

List-I	List-II
A. Barometric levelling	1. Levels at short intervals along a fixed line
B. Trigonometric levelling	2. Difference of atmospheric pressure
C. Reciprocal levelling	3. Level set up near each of two points on opposite banks
D. Profile levelling	4. Vertical angles and measured distances

(a) A-4, B-2, C-3, D-1 (b) A-2, B-4, C-1, D-3 (c) A-2, B-4, C-3, D-1 (d) A-2, B-3, C-4, D-1

28. Assertion (A): Even if the line of collimation of a level is inclined, the difference of level between two points is correct when the back-sight and fore-sight distances are equal.
Reason (R): The errors in the two staff readings are then equal and cancel when one reading is subtracted from the other.
(a) Both A and R are true and R is the correct explanation of A (b) Both A and R are true, but R is not the correct explanation of A
(c) A is true, but R is false (d) Both A and R are false

F. Contours and Digital Elevation Modelling

29. A road is to be aligned at a ruling gradient of 1 in 40 on a contour map of scale 1 cm = 20 m with a contour interval of 2 m. The distance to be stepped off on the map between successive contours is:
(a) 8 cm (b) 2 cm (c) 0.4 cm (d) 4 cm

30. Study the statement and the conclusions and select the most appropriate option.
Statement: On a contour map, the 100 m, 105 m and 110 m contours are found to merge into a single line over a short length.
Conclusion 1: The ground at that place is a vertical cliff.
Conclusion 2: The contour interval of the map changes at that place.
(a) Only conclusion 1 is correct (b) Only conclusion 2 is correct
(c) Both conclusions 1 and 2 are correct (d) Neither conclusion 1 nor 2 is correct

31. Which of the following statements is INCORRECT?

(a) The contour interval is kept constant on a map

(b) Contours cross ridge lines and valley lines at right angles

(c) Indirect methods of contouring are slower but more accurate than the direct method

(d) A TIN represents terrain by a network of triangles

32. S1: LiDAR is an active remote-sensing technique that can give high-resolution digital elevation models.
S2: In a raster (grid) DEM, elevations are stored at the corners of irregular triangles.

(a) Both S1 and S2 are true (b) S1 is true and S2 is false (c) S1 is false and S2 is true (d) Both S1 and S2 are false

G. Tacheometry, EDM and Total Station

33. With a tacheometer, a staff held at 50 m gave an intercept of 0.495 m, and at 100 m an intercept of 0.995 m (horizontal sights). The multiplying and additive constants are:

(a) k = 100, C = 0 (b) k = 100, C = 0.5 m (c) k = 101, C = 0.5 m (d) k = 99, C = 1.0 m

34. A tacheometer (k = 100, C = 0) reads a staff intercept of 1.0 m on a vertical staff with the line of sight inclined at 30° to the horizontal. The vertical component V is:

(a) 86.6 m (b) 75.0 m (c) 43.3 m (d) 50.0 m

35. A subtense bar 2 m long subtends a horizontal angle of 4125″ at the theodolite (1 radian = 206,265″). The horizontal distance to the bar is:

(a) 200 m (b) 50 m (c) 412.5 m (d) 100 m

36. Match List-I with List-II:

List-I	List-II
A. Tellurometer	1. Infrared waves
B. Geodimeter	2. Microwaves
C. Distomat	3. Electronic theodolite, EDM and microprocessor together
D. Total station	4. Visible light

(a) A-2, B-4, C-1, D-3 (b) A-4, B-2, C-1, D-3 (c) A-2, B-1, C-4, D-3 (d) A-1, B-4, C-2, D-3

37. Study the statement and the conclusions and select the most appropriate option.
Statement: A total station is set up over a station and sighted to a prism held on a point.
Conclusion 1: The instrument directly measures the horizontal distance and the difference of elevation.
Conclusion 2: The coordinates shown on the display are computed by the on-board processor from the measured angles and the slope distance.

(a) Only conclusion 1 is correct (b) Only conclusion 2 is correct
(c) Both conclusions 1 and 2 are correct (d) Neither conclusion 1 nor 2 is correct

H. Triangulation and Adjustment

38. In a triangle ABC of a triangulation scheme, the base AB = 1000 m, angle A = 60° and angle B = 90°. The length of side BC is:

(a) 577 m (b) 866 m (c) 1732 m (d) 2000 m

39. The angles A, B and C of a triangle were measured with weights 1, 2 and 2 respectively, and their sum is 180°00′08″. The correction to angle A is:

(a) −1.6″ (b) −2.67″ (c) −2″ (d) −4″

40. Which of the following statements is INCORRECT?

(a) In triangulation all the sides are measured directly and the angles are computed

(b) In primary triangulation the average triangle closure is less than 1″

(c) The base line is measured with the highest possible precision

(d) A satellite station is used when the true station cannot be occupied by the instrument

I. Plane Tabling, Photogrammetry and Map Substitutes

41. Vertical photographs at a scale of 1 : 10,000 are required of ground whose mean elevation is 500 m above datum. If the focal length of the camera is 20 cm, the flying height above datum should be:

(a) 1500 m (b) 2000 m (c) 3000 m (d) 2500 m

42. On a vertical photograph taken from 1500 m above the base of a chimney, the image of its top is 100 mm from the principal point and the relief displacement is 4 mm. The height of the chimney is:

(a) 30 m (b) 37.5 m (c) 120 m (d) 60 m

43. Study the statement and the conclusions and select the most appropriate option.

Statement: Aerial photographs of size 23 cm × 23 cm are taken at a scale of 1 : 20,000 with 60 % longitudinal overlap and 30 % side overlap.

Conclusion 1: The ground area covered by one photograph is 21.16 km².

Conclusion 2: The net new ground area covered by each photograph is about 5.92 km².

(a) Only conclusion 1 is correct (b) Only conclusion 2 is correct

(c) Both conclusions 1 and 2 are correct (d) Neither conclusion 1 nor 2 is correct

44. Consider the following statements about plane tabling:

I. Resection locates the position of the station occupied by the plane table.

II. The three-point problem cannot be solved when the station lies on the circle passing through the three points.

III. Bessel's method is a graphical solution of the three-point problem.

IV. The radiation method needs two instrument stations.

Which of the above statements are correct?

(a) I and II only (b) II and III only (c) I, II and III only (d) I, II, III and IV

45. Match List-I with List-II:

List-I	List-II
A. Mosaic	1. Tilt and relief displacement removed
B. Orthophoto	2. Two overlapping photographs viewed together for depth
C. Photomap	3. Overlapping photographs joined into one picture
D. Stereo pair	4. Photograph with grid lines and place names added

(a) A-3, B-4, C-1, D-2 (b) A-3, B-1, C-4, D-2 (c) A-1, B-3, C-4, D-2 (d) A-2, B-1, C-4, D-3

J. Remote Sensing, GPS and Field Astronomy

46. A satellite sensor records data with a radiometric resolution of 11 bits. The number of grey levels it can record is:

(a) 2048 (b) 256 (c) 1024 (d) 4096

47. For a pixel, the reflectance is 0.50 in the near-infrared band and 0.10 in the red band. Its NDVI is:

(a) 0.67 (b) 0.40 (c) 0.80 (d) 1.50

48. Which of the following statements about GPS is INCORRECT?

(a) At least four satellites are needed for a three-dimensional position fix

(b) In DGPS a reference receiver is placed on a point of known position

(c) GPS positions are referred to the WGS-84 datum

(d) GPS satellites are placed in geostationary orbit at about 36,000 km

49. At a place of latitude 30° N, a star has a declination of 70° N. Its altitude at lower culmination is:

(a) 50° (b) 40° (c) 10° (d) 20°

50. Place A is on longitude 75° E and place B on longitude 90° E. When the local mean time at A is 08:00, the local mean time at B is:

(a) 07:00 (b) 09:00 (c) 08:15 (d) 07:45

Answer key with short solutions

A. Fundamentals, Scales and Errors

1. (a) Both are true and R is the reason: an arc of about 12 km is only about 1 cm longer than its chord, so areas up to about 250 km² are treated as plane.

2. (c) Shrinkage factor = 9.6/10 = 0.96. Original plan area = 92.16/0.96² = 100 cm². Ground area = 100 × (20 m)² = 40,000 m² = 4 hectares. (3.69 ha ignores the shrinkage.)

3. (a) Cadastral → property boundaries (revenue maps); hydrographic → water bodies; geodetic → curvature considered; reconnaissance → first rough inspection.

4. (d) 1 cm = 250 m, so 1 cm² = 62,500 m². 16 × 62,500 = 10,00,000 m² = 1 km². For areas, use the square of the scale.

B. Chain and Tape

5. (c) True area = $(L'/L)^2$ × measured area = $(29.85/30)^2$ × 10,000 = 0.995^2 × 10,000 = 9,900.25 m². A short chain over-measures. (9,950 forgets to square the ratio.)

6. (b) $C = W^2L/(24P^2) = 12^2 \times 30/(24 \times 100^2) = 4320/2{,}40{,}000 = 0.018$ m, always negative. W is the total weight of the tape between the supports.

7. (c) Correction = $\alpha(T - T_0)L = 12 \times 10^{-6} \times 20 \times 600 = +0.144$ m. A warm tape is longer, so it under-measures; the correction is added.

8. (a) A chain that is too long fits fewer times into the line, so the recorded length is too small and the correction is positive. Conclusion 2 is wrong.

9. (b) S1 is the definition of normal tension. S2 is false: invar (nickel–steel alloy) is chosen for its very low coefficient of thermal expansion.

10. (b) Across a pond or river the far station can be seen (ranging is free) but the chain cannot be laid. A hill obstructs ranging only; a building or forest obstructs both.

C. Compass Surveying

11. (d) Back bearing of AB = 75° + 180° = 255° (direction BA). Angle ABC = bearing of BA − bearing of BC = 255° − 160° = 95°. (85° is the deflection angle.)

12. (d) PQ: difference exactly 180°, so P and Q are free. Correct BB of QR = 200°15′ − 180° = 20°15′; observed 21°45′, so R reads 1°30′ too much. Corrected FB of RS = 285°00′ − 1°30′ = 283°30′.

13. (a) True bearing = magnetic bearing − west declination, so magnetic bearing = 210° + 4° = 214°. (With west declination the magnetic north lies west of true north, so magnetic bearings are larger.)

14. (c) 'Gonic' goes with declination: isogonic = equal, agonic = zero. 'Clinic' goes with dip (inclination): isoclinic = equal, aclinic = zero.

15. (c) A is true (the prism lets the observer read while sighting). R is false: in a prismatic compass the ring is attached to the needle and stays still; a ring fixed to the box is the surveyor's compass.

D. Theodolite, Traversing and Coordinates

16. (b) Latitude = 100 × 0.8 = 80 m south (−); departure = 100 × 0.6 = 60 m east (+). B = (200 − 80, 100 + 60) = N 120 m, E 160 m.

17. (d) Correction = − total error × length of side / perimeter = −0.40 × 200/800 = −0.10 m. The sign is opposite to the error.

18. (c) ΔN = −100 m (south), ΔE = +100 m (east) → reduced bearing S 45° E → WCB = 180° − 45° = 135°. Fix the quadrant from the signs before using tan.

19. (b) III is false: an open traverse does not return to a known point, so it has no closing error and no such check. The other three are standard results.

20. (b) Transiting (plunging, reversing) is rotation about the horizontal axis. Swinging is rotation about the vertical axis. Changing face needs both: transit, then swing through 180°.

21. (a) Reading both verniers removes eccentricity. Both-face readings remove collimation and horizontal-axis errors, but not the error of a vertical axis that is not truly vertical — only careful levelling does.

E. Levelling

22. (a) HI = 50.000 + 1.250 = 51.250 m. RL of C = 51.250 − 0.940 = 50.310 m. The middle reading is an intermediate sight and does not affect C.

23. (d) HI = 101.500 m. An inverted staff reading is negative, so RL = HI − (−2.500) = 104.000 m: the beam is above the line of sight.

24. (d) Last RL − first RL = ΣBS − ΣFS = 6.520 − 4.870 = +1.650 m → 151.650 m.

25. (a) Add the two horizon distances: 3.855(√64 + √9) = 3.855 × 11 = 42.4 km. (32.9 km wrongly uses √(64 + 9).)

26. (b) α = s/(nD) = 0.05/(5 × 100) = 1 × 10^{-4} rad = 20.6″. A smaller value per division means a more sensitive tube.

27. (c) The same idea as 2023 Q132: barometric → pressure; trigonometric → vertical angles and distances; reciprocal → both banks; profile → along a fixed line.

28. (a) Collimation error is proportional to distance. With equal distances both readings are wrong by the same amount, so their difference is right. Curvature and refraction cancel in the same way.

F. Contours and Digital Elevation Modelling

29. (d) Horizontal distance for a 2 m rise at 1 in 40 = 80 m. On the map, 80/20 = 4 cm. This is how a contour gradient is traced.

30. (a) Contours of different values join only where the ground is vertical. The contour interval stays constant over the whole map.

31. (c) It is the other way round: the direct method (tracing each contour on the ground) is slower and more accurate; indirect methods (grid, cross-sections, tacheometry) are faster and less accurate.

32. (b) S1 is true: LiDAR sends its own laser pulses. S2 describes a TIN; a raster DEM stores one elevation per square cell of a regular grid.

G. Tacheometry, EDM and Total Station

33. (b) D = ks + C. Subtract: 50 = k × 0.5 → k = 100. Then 50 = 100 × 0.495 + C → C = 0.5 m. An anallactic lens would make C zero.

34. (c) V = ks sin 2θ / 2 = 100 × 1 × sin 60°/2 = 43.3 m. (75 m is the horizontal distance ks $\cos^2\theta$; 86.6 m forgets the division by 2.)

35. (d) α = 4125/206,265 = 0.02 rad. D = b/α = 2/0.02 = 100 m.

36. (a) Tellurometer → microwaves (long range); Geodimeter → visible light; Distomat → infrared (short range, as in total stations).

37. (b) A total station measures only three things — horizontal angle, vertical angle and slope distance. Horizontal distance, level difference and coordinates are all computed.

H. Triangulation and Adjustment

38. (c) Angle C = 30°. Sine rule: BC/sin A = AB/sin C → BC = 1000 × 0.866/0.5 = 1732 m. (2000 m is side AC, opposite the 90° angle.)

39. (d) Corrections are in inverse proportion to the weights: 1 : ½ : ½. Total 8″ → A gets 8 × 1/2 = 4″, B and C get 2″ each, all negative. The least reliable angle takes the most.

40. (a) That describes trilateration. In triangulation all the angles and only one base line are measured; the other sides are computed by the sine rule.

I. Plane Tabling, Photogrammetry and Map Substitutes

41. (d) Scale = f/(H − h) → H − h = 0.2 × 10,000 = 2000 m → H = 2000 + 500 = 2500 m. (2000 m is the height above the ground, not above datum.)

42. (d) d = rh/H → h = dH/r = 4 × 1500/100 = 60 m.

43. (c) Ground side = 0.23 × 20,000 = 4.6 km → 4.6^2 = 21.16 km^2. Net = (0.4 × 4.6) × (0.7 × 4.6) = 1.84 × 3.22 = 5.92 km^2. Both are correct — work the numbers before deciding (2023 Q139 gave wrong numbers).

44. (c) IV is false: radiation is done from a single station, with every distance measured. Two stations at the ends of a base are needed for intersection.

45. (b) Mosaic → photographs joined; orthophoto → uniform scale after removing tilt and relief displacement; photomap → mosaic with map information; stereo pair → 3-D viewing.

J. Remote Sensing, GPS and Field Astronomy

46. (a) Grey levels = 2^n = 2^{11} = 2048. (8 bits = 256, 10 bits = 1024, 12 bits = 4096.)

47. (a) NDVI = (NIR − Red)/(NIR + Red) = 0.40/0.60 = 0.67 — a high value, typical of healthy green vegetation. NDVI always lies between −1 and +1.

48. (d) GPS satellites orbit at about 20,200 km in six planes inclined at 55°, with a period of about 12 hours. Geostationary orbit (36,000 km) is used by weather and communication satellites.

49. (c) Altitude at lower culmination = $\varphi + \delta - 90° = 30° + 70° - 90° = 10°$. It stays above the horizon, so the star is circumpolar ($\delta > 90° - \varphi$). 50° is the altitude at upper culmination.

50. (b) Difference of longitude = 15° = 1 hour. B is to the east, so its time is ahead: 09:00.

Score analysis — fill this in and revise the weakest section first

Section	Questions	Max	Correct	Wrong	Net (C − W/3)
A. Fundamentals, Scales and Errors	1–4	4			
B. Chain and Tape	5–10	6			
C. Compass Surveying	11–15	5			
D. Theodolite, Traversing and Coordinates	16–21	6			
E. Levelling	22–28	7			
F. Contours and Digital Elevation Modelling	29–32	4			
G. Tacheometry, EDM and Total Station	33–37	5			
H. Triangulation and Adjustment	38–40	3			
I. Plane Tabling, Photogrammetry and Map Substitutes	41–45	5			
J. Remote Sensing, GPS and Field Astronomy	46–50	5			
Total	1–50	**50**			

Net 35 or more: ready, move to the next module. **25–35:** re-read the weak topics and redo their quizzes. **Below 25:** 3–4 more days on Module 3, then retake after a week.